

# Random Matrices

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**Definition 0.1.** Consider a matrix of random entries:

$$\begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{pmatrix}$$

These entries are random variables. Let  $X_1, \dots, X_n$  be vectors in  $\mathbb{R}^d$ , where each  $X_i = (X_{i1}, \dots, X_{id})$  are random variables.

**Definition 0.2.** The variance of a random variable  $X_{ij}$  is defined as:

$$\frac{1}{n} \sum_{i=1}^n X_{ii}^2 \longrightarrow \text{Variance}$$

The covariance between two random variables  $X_{i1}$  and  $X_{i2}$  is defined as:

$$\frac{1}{n} \sum_{i=1}^n X_{i1} X_{i2} \longrightarrow \text{Covariance}$$

Let  $X$  be the data matrix:

$$X = \begin{pmatrix} X_{11} & \cdots & X_{1d} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{nd} \end{pmatrix}$$

The sample covariance matrix is given by:

$$\frac{XX^\top}{n}$$

**Definition 0.3.** The eigenvalues  $\lambda$  of a matrix  $A$  are defined by the equation:

$$Av = \lambda v \quad \text{for } v \neq 0$$

The trace of a matrix  $A$  is the sum of its eigenvalues:

$$\text{tr}(A) = \sum_{i=1}^n A_{ii} = \sum_{i=1}^n \lambda_i$$

*Remark 0.1.* The set of eigenvalues of a random matrix is called the empirical spectral distribution:

$$\frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i}$$

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Consider a random matrix  $A = \begin{pmatrix} A_{11} & A_{1j} & \cdots \\ \vdots & \vdots & \ddots \\ A_{m1} & A_{mj} & \cdots & A_{mn} \end{pmatrix}$  where the entries are  $\pm 1$  with probability  $\frac{1}{2}$ .

Note that  $A_{ji} = A_{ij}$ .

Trace properties:

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i,$$

$$\text{tr}(A^2) = \sum_i A_{ii}^2 = \sum_i \lambda_i^2,$$

$$\text{tr}(A^k) = \sum_i A_{ii}^k = \sum_i \lambda_i^k.$$

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